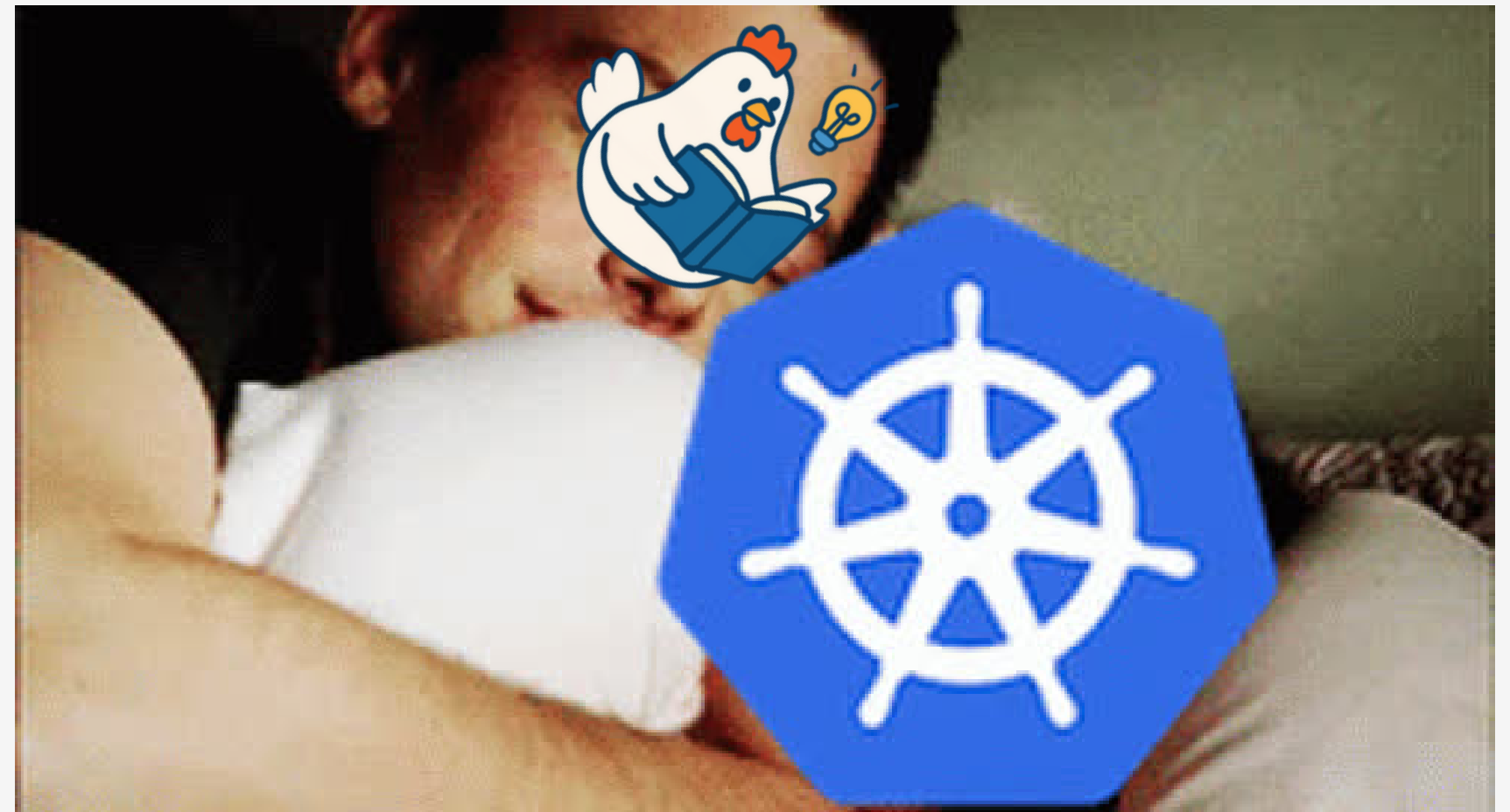


Moojidle-k8s (A better Moodle via Kubernetes)



資訊四 111703004 林子齊
資訊四 111703044 劉子宏
資訊四 111703040 游宗諺

資訊四 111703003 廖經翔
資訊四 111703022 黃鈺盛
資訊三 112703016 劉宸均

<https://moojidle-k8s.online>

課程邀請碼：OADLN2



MOOJIDLE

實用意義與創新性

實用意義與創新性

不知道自己還有哪些作業沒交

找好久才找到助教與老師的聯絡方式

課程中找不到一個公開討論的平台

作業不管有沒有過期都標示「已經過期」

要滑很久才能找到當周的資料

網頁充滿著不曾用過的區塊

功能面

使用者管理

- 註冊 / 登入 / 登出
- 角色分類（學生／教師／助教）
- 編輯個人資料、設定標籤（tag）

通知中心

- 課程異動通知
- 新留言/公告通知
- 作業與活動提醒

課程資料管理

- 教師可新增、編輯、刪除課程
- 學生可瀏覽已開課程、查詢詳細資訊
- 上傳與下載課程教材
- 課程內學生分組
- 發布作業、設定繳交期限
- 學生上傳作業
- 評分與回饋
- 改作業模式
- 考試上傳、考試評分

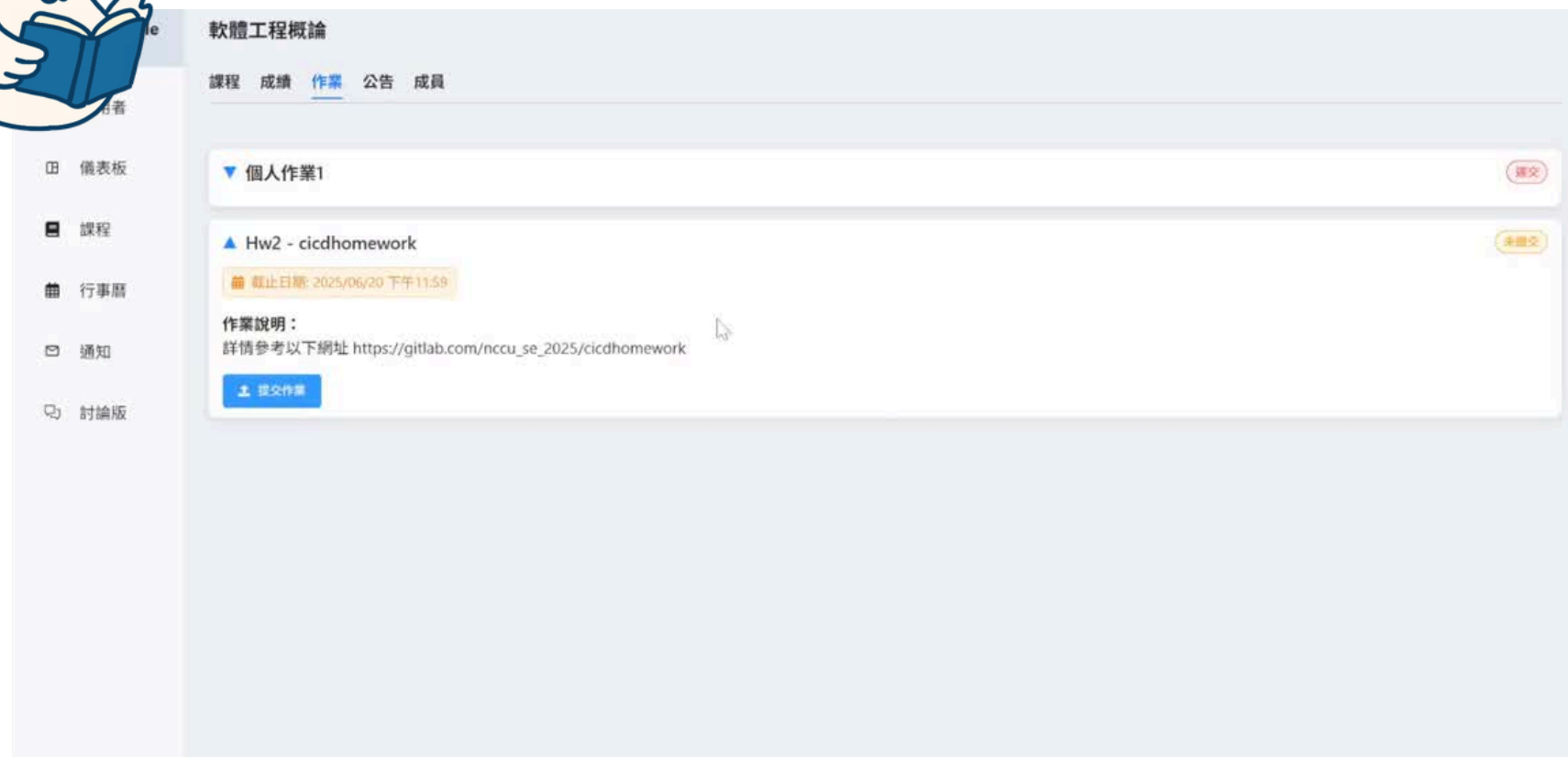
行事曆

- 顯示所有課程相關事件
- 每月、每週與每日檢視模式

討論版

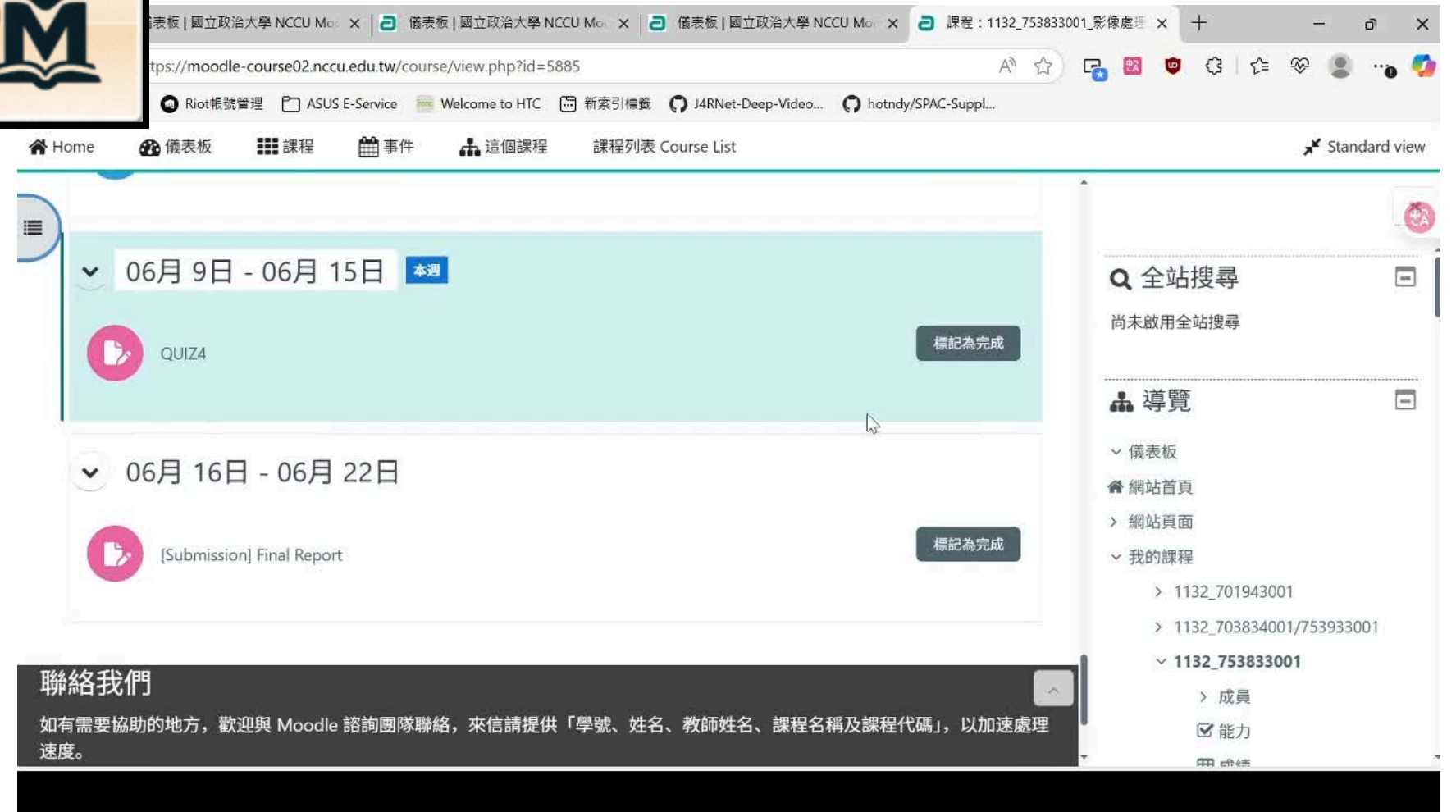
- Course -> Discussion -> post
- 課程內、外討論版
- Markdown

學生想要找到課程中某個未完成的作業



5 sec 單次跳轉

節省 55% 時間



11 sec 多次跳轉

(會隨著課程資料增多而加長時間)

找到最後一周的教材



Moojidle

物件導向程式設計

課程 成績 作業 公告 成員

Week	Lecture	Assignments	Exams
1 2/16 - 2/22	01-Introduction (2025/2/17) Assignment繳交狀況 (2025/2/17)	Assignment0	
2 2/23 - 3/1	02-better-c (2025/2/24) Makefile 語法和示範 (2025/2/24)		
3 3/2 - 3/8	03-function-and-reference (2025/3/3)	Assignment1	
4 3/9 - 3/15	04-class (2025/3/10)	Assignment2	
5 3/16 - 3/22	05-more-on-class (2025/3/17) CPE 大學程式能力檢定 (2025/3/17)		Quiz 1
6 3/23 - 3/29	06-operator-overloading (2025/3/24)	Assignment3	
7 3/30 - 4/5	The Software Engineering Identity Crisis (2025/3/31)		
8 4/6 - 4/12			期中考試
9 4/13 - 4/19	07-inheritance (2025/4/14) Awesome C++ (2025/4/14)	Assignment4	
10 4/20 - 4/26	08-advanced-inheritance (2025/4/21)		

To Do

- Assignment1 提交
截止：2025/3/11 下午11:59
- Assignment0 提交
截止：2025/2/25 下午11:59
- Assignment2 提交
截止：2025/3/25 下午11:59
- Assignment3 提交
截止：2025/4/8 下午11:59
- Assignment4 提交
截止：2025/4/22 下午11:59
- Assignment5 提交
截止：2025/5/16 下午11:59

9.6 sec

節省 42% 時間



1132_701943001_機器學習_...

Home 儀表板 課程 事件 這個課程 課程列表 Course List

Standard view

站搜尋

一般

- 公告 Announcements
- 1132_701943001_機器學習_Machine Learning_課程大綱_course outline
- 期末報告分組名單

全部收合

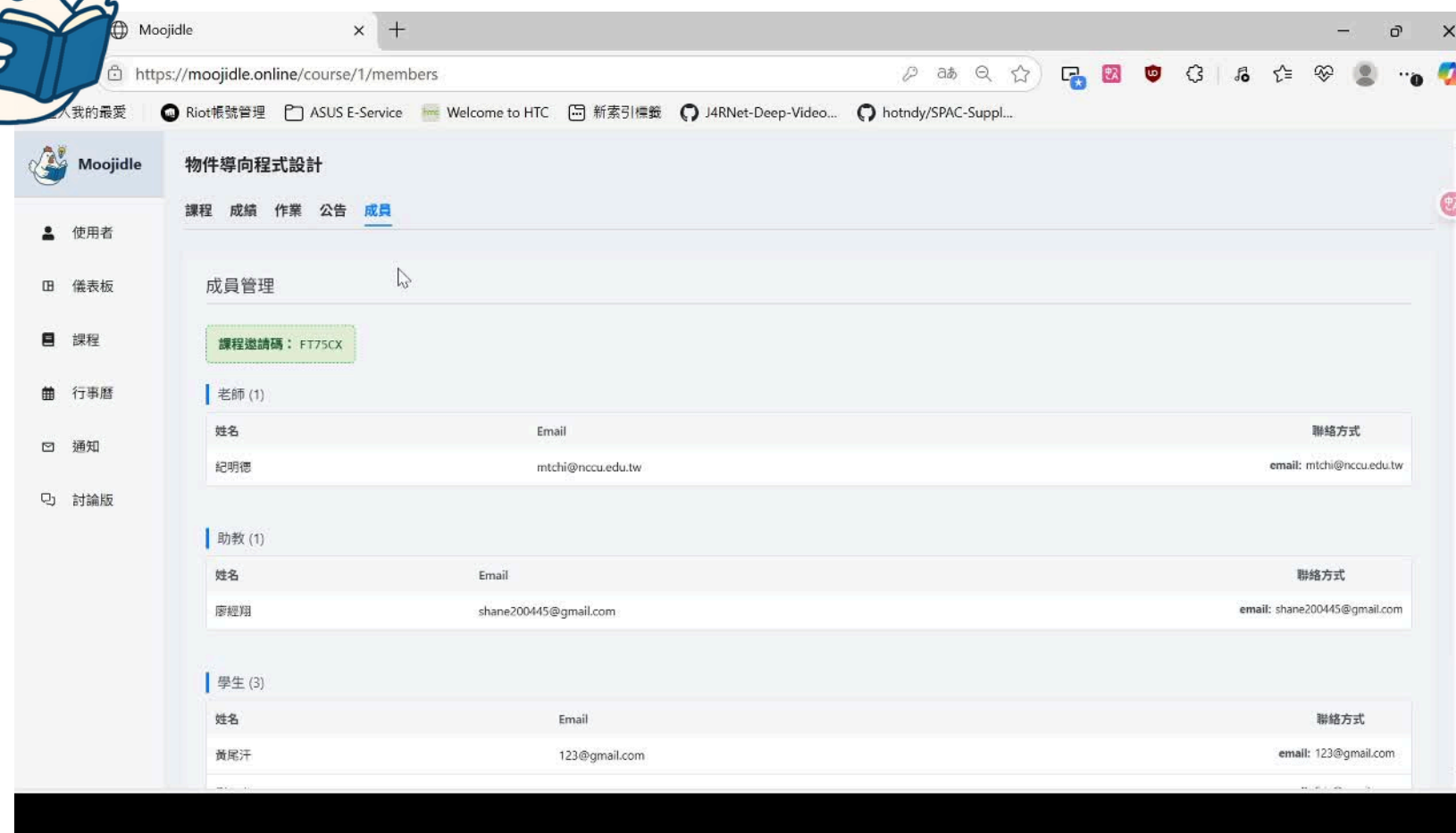
1132_701943001

成員

16.6 sec

(會隨著課程資料增多而加長時間)

尋找助教、老師的聯絡方式



12 sec

節省 71% 時間



42 sec

考試/作業評分



Moojidle

https://moojidle.online/course/1/exams

登入我的最愛

Riot帳號管理

ASUS E-Service

Welcome to HTC

新索引標籤

J4RNet-Deep-Video...

hotndy/SPAC-Suppl...

Moojidle

物件導向程式設計

課程 成績 作業 考試 公告 成員

Quiz 1 期中考試 Quiz 2 期末上機考

期中考試

實體紙筆考試
考試地點 分兩間教室
學號尾單數到 綜合院館 (北棟) 綜北270203
學號尾雙數到 綜北270204
不可使用任何計算或連網設備。

展開

Student Name	Grade (0-100)	Status	Actions	GradeForm
劉子宏	Not graded	待評分	Grade	
林子齊	Not graded	待評分	Grade	
dummy2	Not graded	待評分	Grade	
dummy4	Not graded	待評分	Grade	

37 sec

節省 47% 時間



自2025年02月19日,19:00起即可用

02月24日 - 03月2日

Week 2: cin/cout, string, Coding Style, git basic

slide: lab1

自2025年02月26日,19:00起即可用

03月3日 - 03月9日

Week 3: Array & Pointer, Pass Argument, STL

slide: Lab2

自2025年03月5日,19:00起即可用

03月10日 - 03月16日

Week 4: Class, Object, UML

全站搜尋

尚未啟用全站搜尋

導覽

儀表板

網站首頁

網站頁面

我的課程

1132_703834001/753933001

1132_703053001/753873001

1132_753806001/703817001

1132_703955001

成員

能力

成績

一般

02月17日 - 02月23日

02月24日 - 03月2日

03月3日 - 03月9日

03月10日 - 03月16日

03月17日 - 03月23日

03月24日 - 03月30日

03月31日 - 04月6日

04月7日 - 04月13日

04月14日 - 04月20日

12.8 sec

截止日期: 2025年03月26日,23:59

62分36 重設表格偏好

繳交作業

沒有繳交作業

尚未評分

已經超過繳交時間: 77 日 10 小時

學生可以編修這已繳交的作業

評論 (0)

成績

得分 (配分100)

目前成績單中的分數

尚未評分

評語回饋

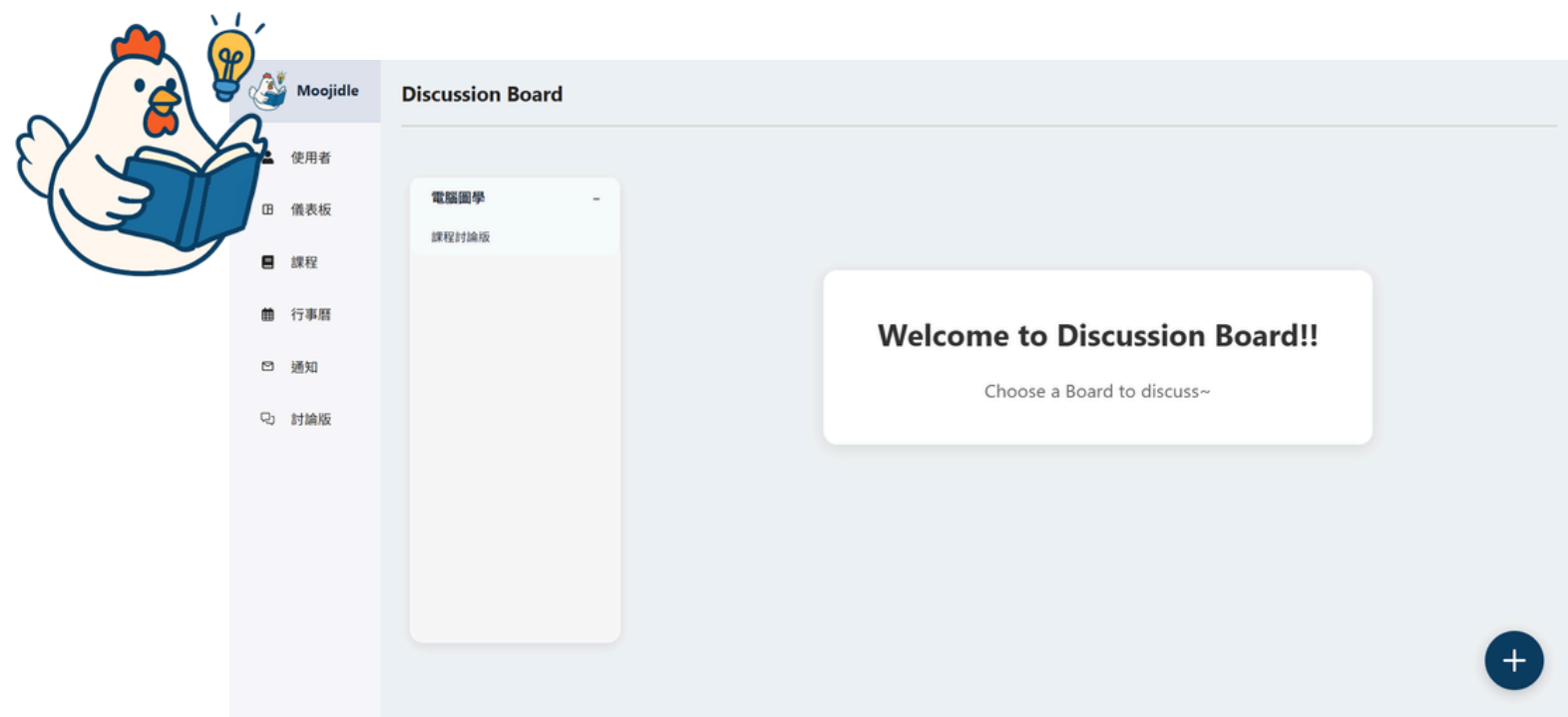
Edit View Insert Format Tools Table Help

0 words tiny

通知學生 儲存變更 儲存後顯示下一位 結束

56.3 sec

討論版



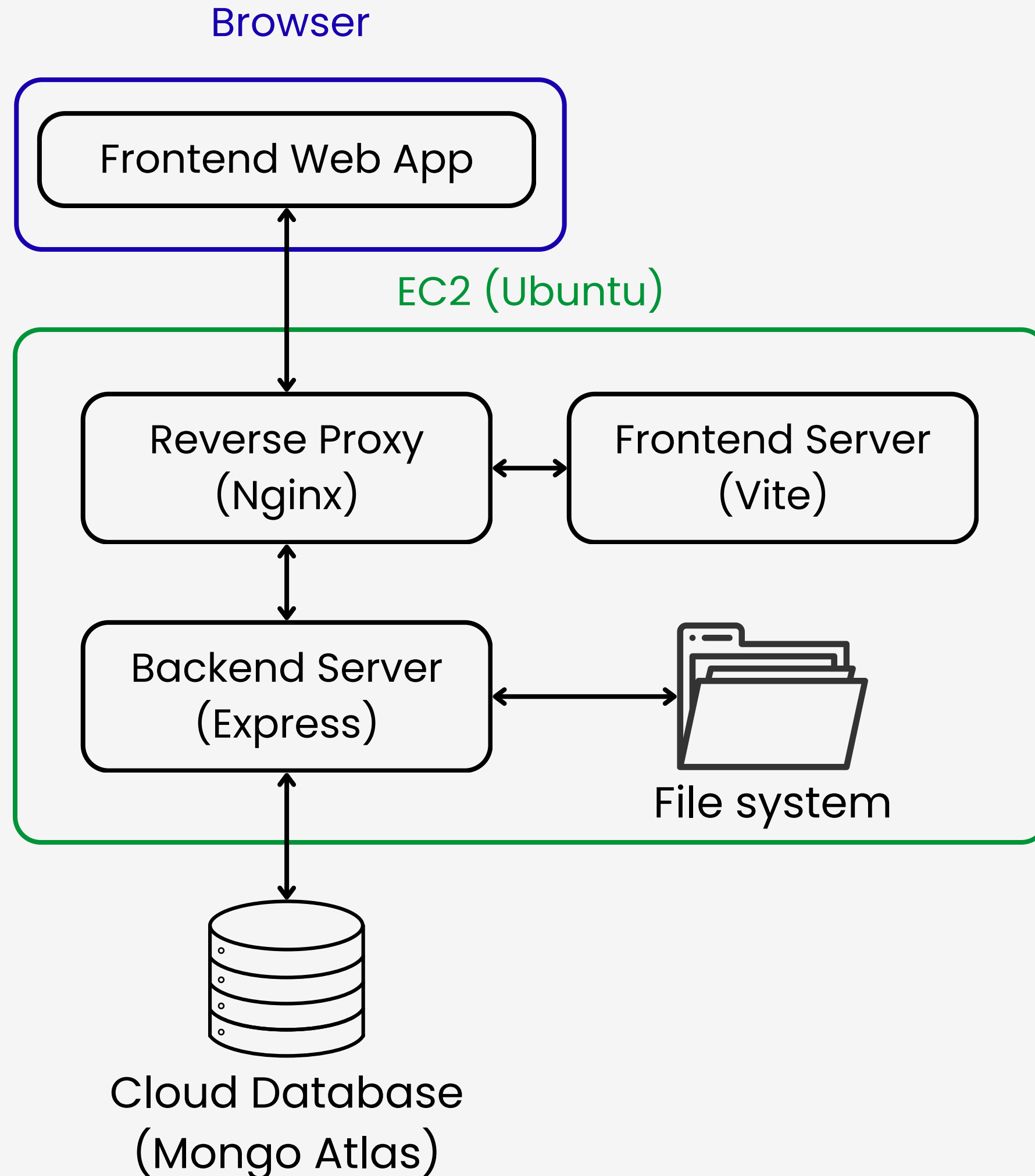
一般型討論區

討論區	說明	議題	已經訂閱	電子郵件摘要型式
公告 Announcements	一般消息與公告	6	是	預設(沒有摘要) ▼
討論區	歡迎提出問題與協助同學解答	1	否	預設(沒有摘要) ▼





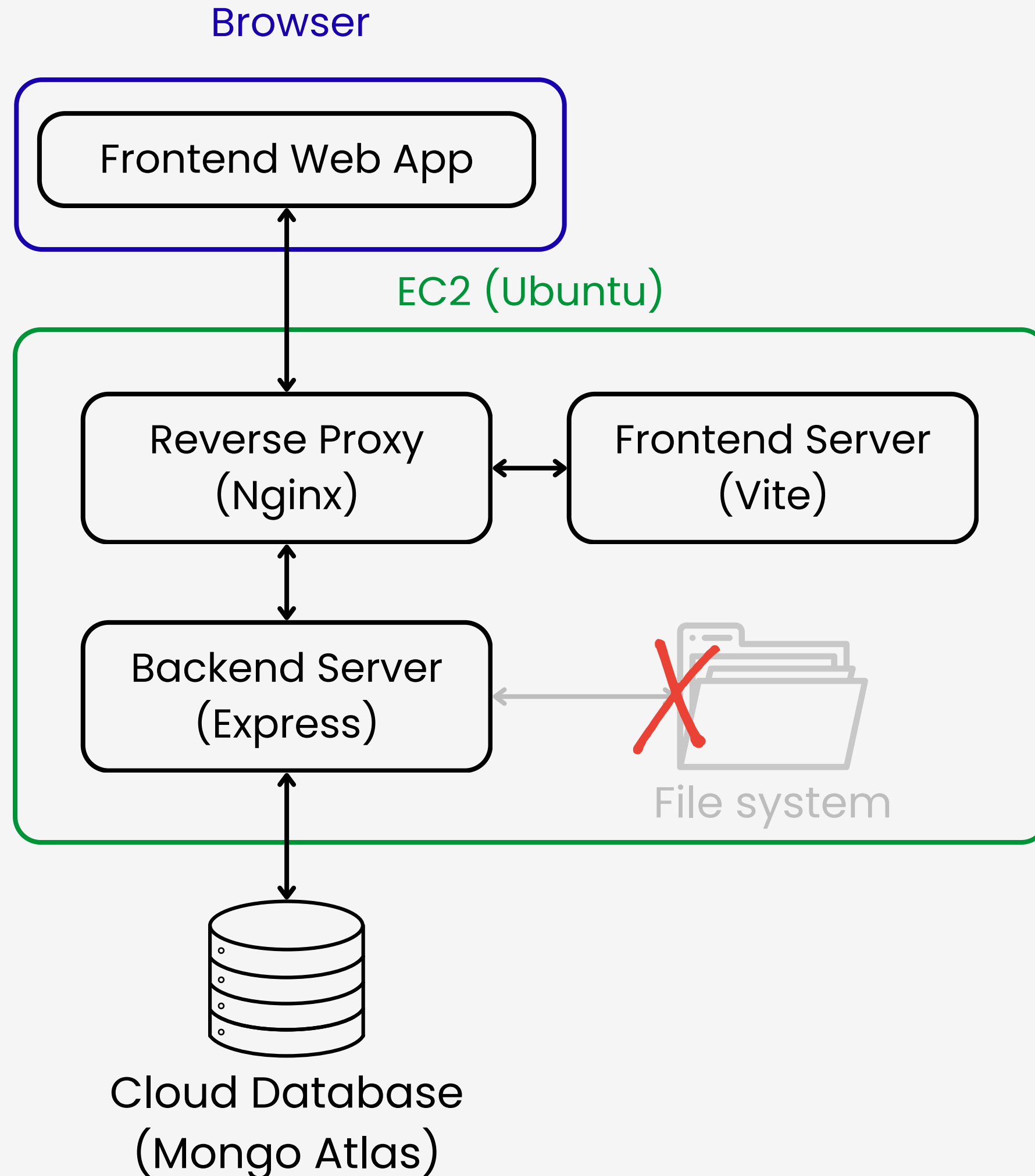
專案架構演進



本課程的起點

最原始的 monolith 架構，**直接在 EC2 的 OS (Ubuntu) 上執行**。

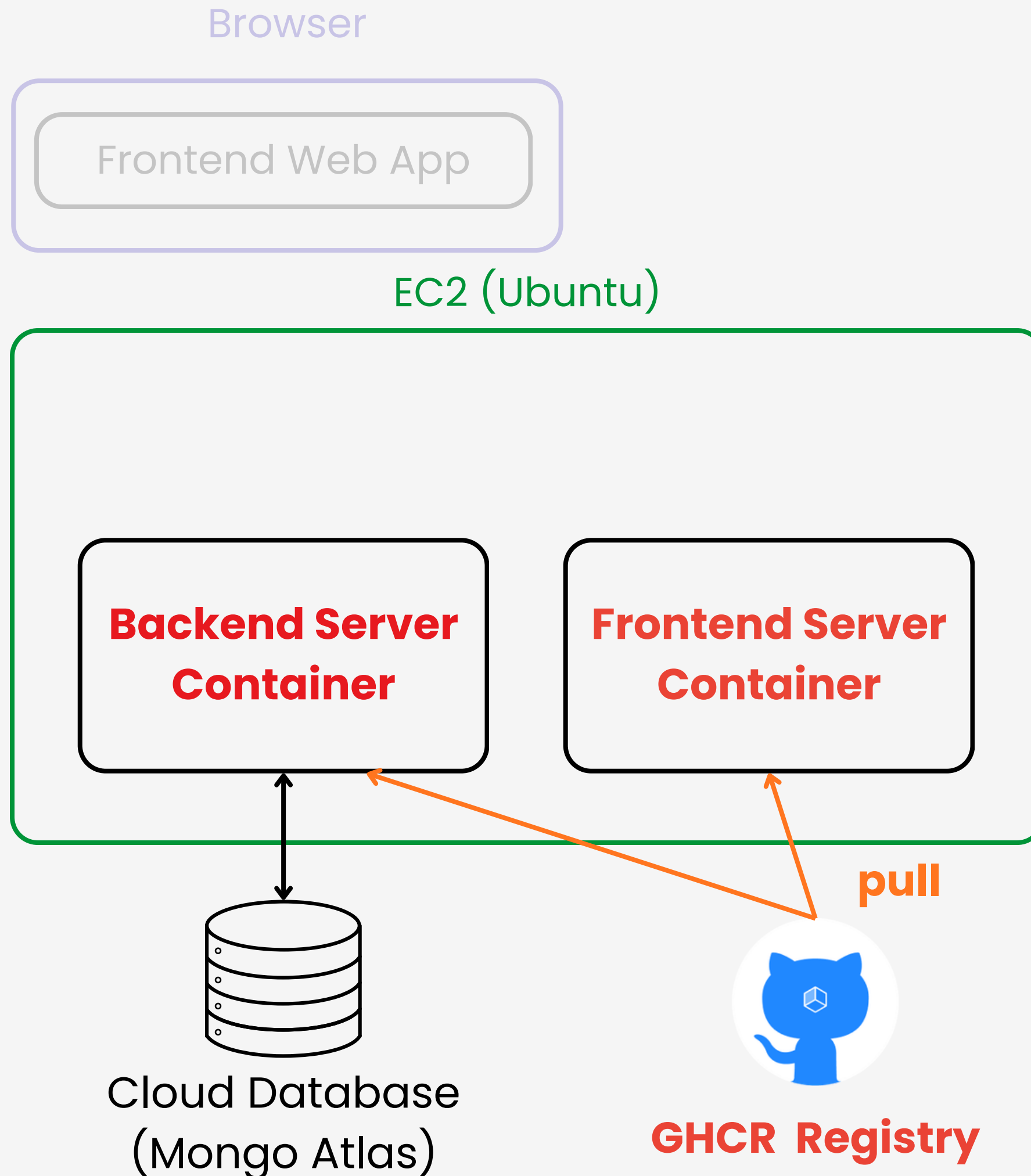
- nginx 作為 reverse proxy，將流量轉發給 frontend server、backend server
- frontend 以 vite production server 對外服務 (不是以靜態檔案部署)
- 檔案儲存直接寫入本機磁碟
- 資料庫使用外部的 MongoDB Atlas。
- 整個架構沒有容器化，且檔案直接儲存在本機磁碟(**stateful**)，**無法輕易水平擴展**。



Storage Decouple

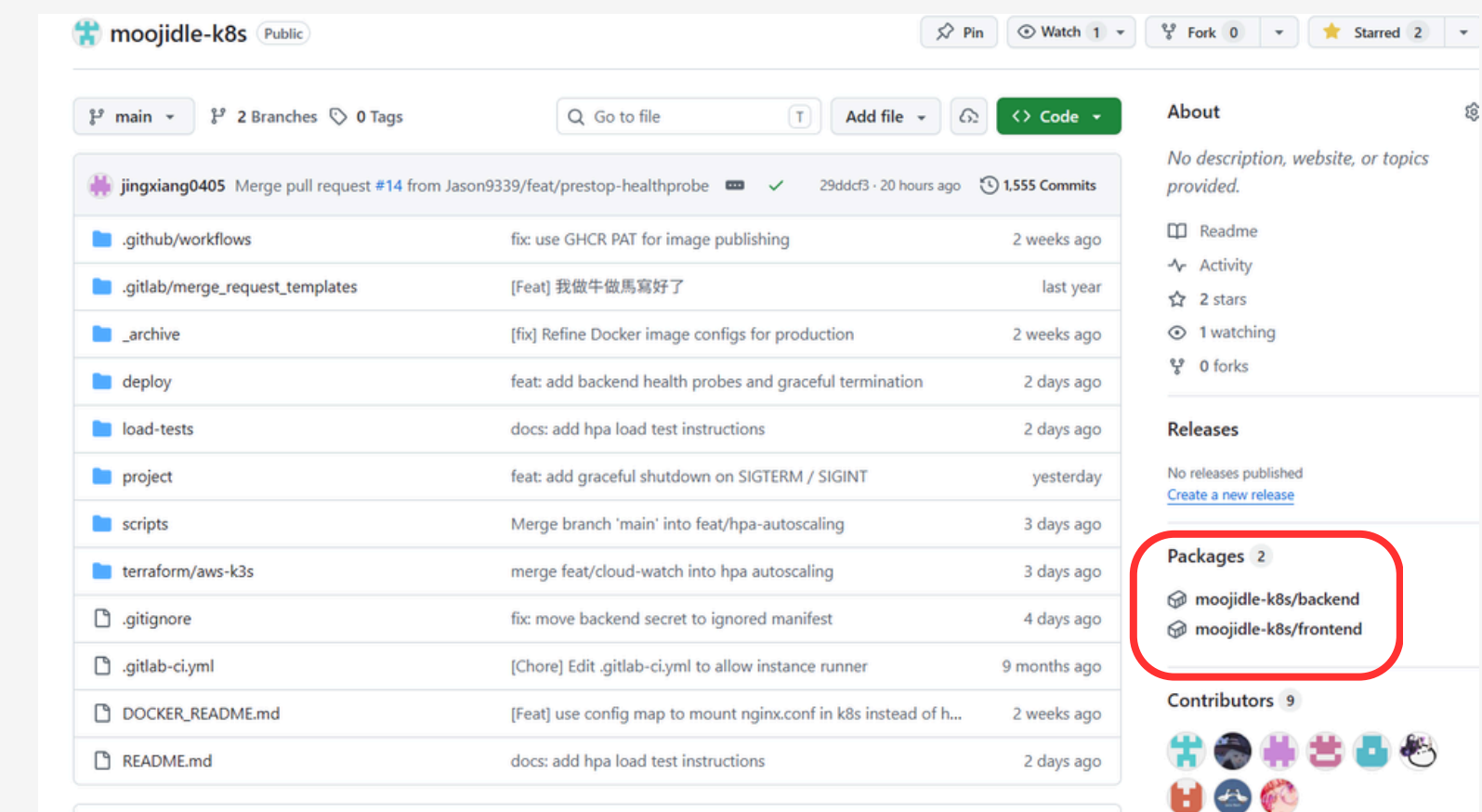
檔案儲存從本機磁碟移至 MongoDB GridFS

Stateful → Stateless



Containerization

將前後端服務容器化，並透過 GitHub Actions 自動建置 image，再由 Kubernetes 從 GHCR 拉取 image 進行部署與更新。

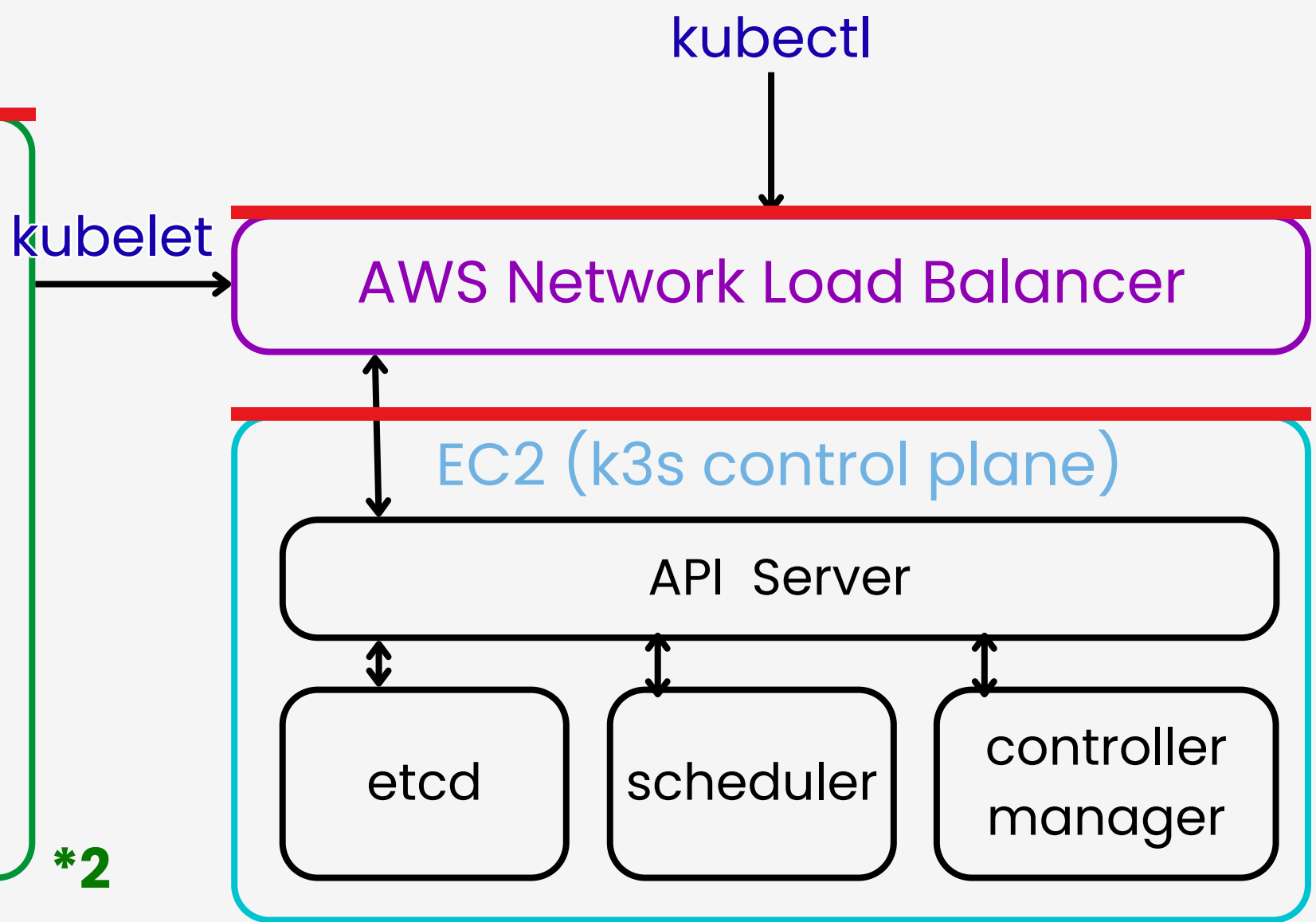
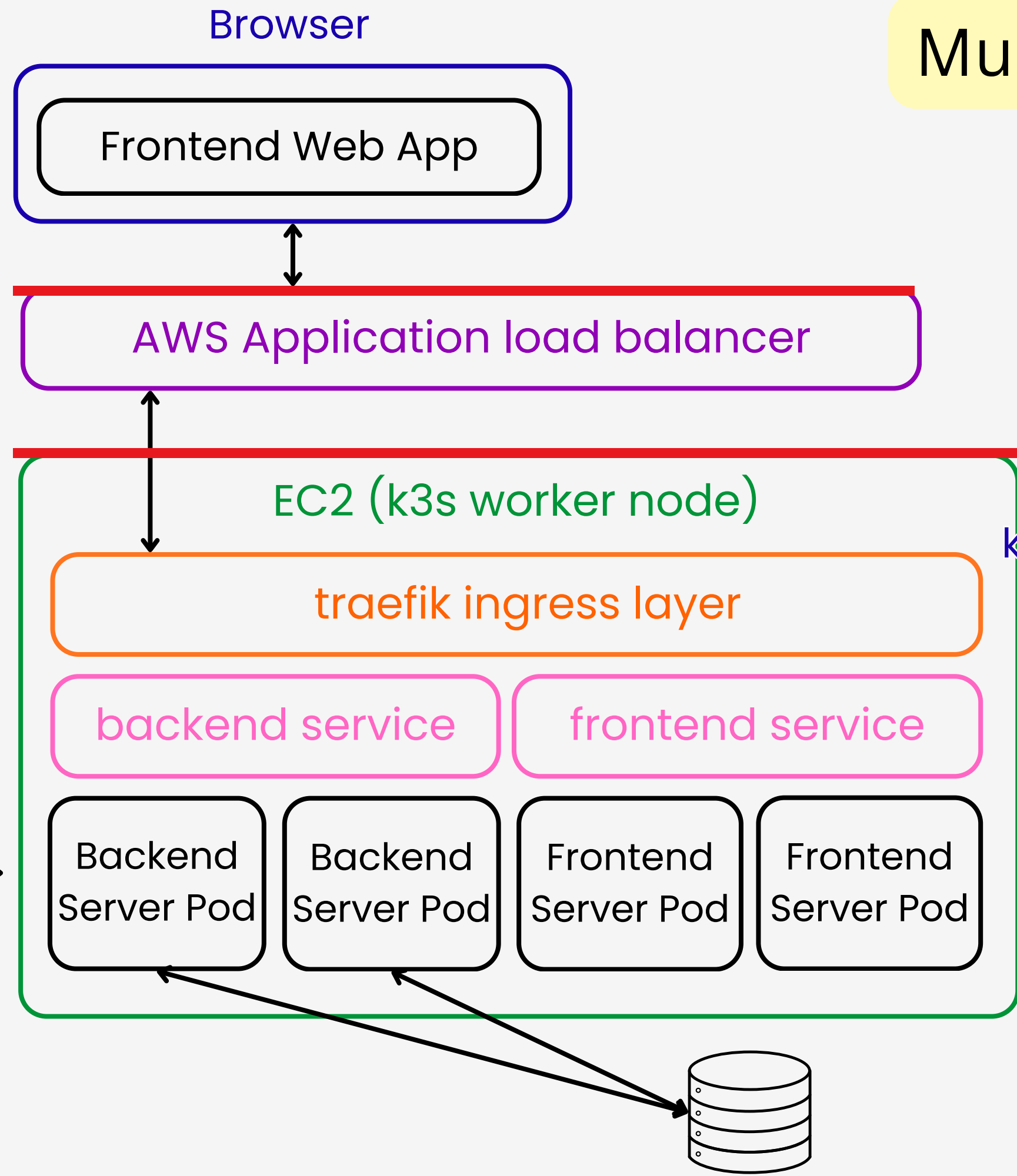


Multi-Node K3s Cluster on AWS (Manuscript)

手動完整部署約需 0.5 - 1 小時

Security Group

GHCR Registry



kubelet

kubectl

Browser

Frontend Web App

Cloudflare (Reverse Proxy)

AWS Application load balancer



EC2 (k3s worker node)

traefik ingress layer

backend service

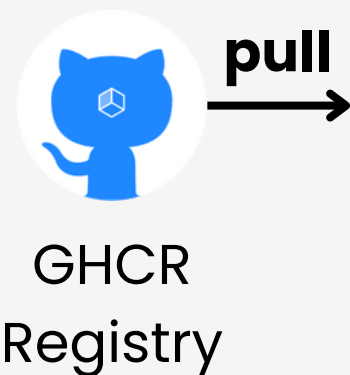
frontend service

Backend
Server Pod

Backend
Server Pod

Frontend
Server Pod

Frontend
Server Pod



pull

Cloud Database

DNS + TLS

DNS 與 TLS 憑證整合至
AWS 雲端服務管理

kubectrl

AWS Network Load Balancer

EC2 (k3s control plane)

API Server

etcd

scheduler

controller
manager

kubelet

*3

*2

Request Flow

Request 如何從 Browser 走到 Kubernetes

Browser

Frontend Web App

DNS query moojield-k8s.online
return Cloudflare Proxy IPs

DNS Resolver

HTTPS

【TLS #1】 Issuer: CloudFlare CA / Let's Encrypt

Cloudflare (Reverse Proxy)

Browser Handshake with CloudFlare Proxy
establish first session key
with CloudFlare's Certificate

【TLS #2】 Issuer: AWS ACM CA

Cloudflare Handshake with AWS Load Balancer
establish second session key
with ACM's Certificate

HTTPS

AWS Application load balancer

NO TLS

HTTP

EC2 (k3s worker node)

traefik ingress layer

NO TLS

HTTP

EC2 (k3s worker node)

traefik ingress layer

Ingress rule
看網址:

/api/

/

rewrite to

/

:3000

/

:8080

backend service

frontend service

:3000

:3000

:8080

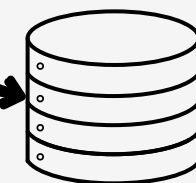
:8080

Backend
Server Pod

Backend
Server Pod

Frontend
Server Pod

Frontend
Server Pod



Cloud Database

EC2 (k3s control plane)

traefik ingress layer

EC2 (k3s worker node)

traefik ingress layer

backend
service

frontend
service

Backend
Server
Pod

Backend
Server
Pod

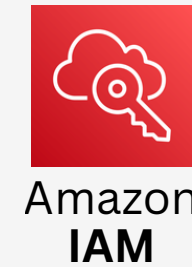
Frontend
Server
Pod



專案應用技術

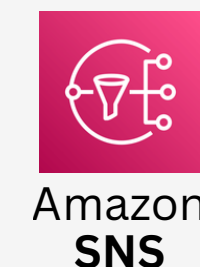
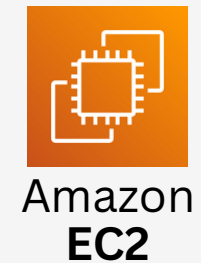
皆為本堂課使用，不含舊專案中實作的相關技術

專案應用技術



Cloud Infrastructure / AWS

- EC2：AWS 提供的虛擬機器服務。
- Security Group：AWS 虛擬防火牆，用於控制網路流量的進出規則。
- NLB：AWS 第四層（L4）Load Balancer，讓本地 kubectl 可以連到 cluster。
- ALB：AWS 第七層（L7）Load Balancer，作為 Web App 的對外入口。
- ACM：AWS 憑證管理服務，用於簽發與管理 SSL/TLS 憑證。
- CloudWatch：AWS 監控服務，用於蒐集與分析系統及應用程式指標。
- SNS：AWS 訊息通知服務，支援 Email、SMS 與事件推送。



專案應用技術



Kubernetes (k3s)

- Deployments : 以 declarative 的方式管理 pod
 - replicas : 幾個 pod
 - containers : 什麼 pod
 - resources : pod 多強
 - probes : pod 多健康
 - lifecycle : pod 死前動作
- Services : 全部型別皆設為外面看不到的 ClusterIP (traefik Ingress Service 除外)
- ConfigMaps: 前後端環境變數設定 (前端: nginx.conf + 後端: port)
- Secrets : MongoDB Atlas 登入存取帳密
- Ingress : 透過 /api 前綴來分流
- HorizontalPodAutoscaler : 配合 Deployments 的 resources 自動擴展 pod



Infrastructure as Code

- Terraform (宣告與建置)：建立 AWS Infra、CloudFlare、MongoDB Atlas IP whitelist
- Bash Script (流程控制與部署)：執行 TF、打通 SSH、設定 Kubeconfig、呼叫 Kubectl

Testing / Performance Validation

- k6：進行 smoke test、stress test、HPA 壓力測試。

CI/CD & Image Registry

- Docker：將 frontend/backend containerize。
- GHCR：儲存 Docker images。
- GitHub Actions：自動 build and push frontend/backend images。

Database

- MongoDB Atlas：雲端資料庫服務。

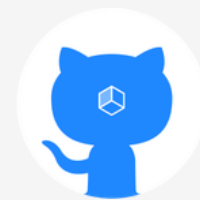
Infrastructure as Code (IaC)

terraform



kubectl

kubelet



pull

GHCR
Registry

Frontend Web App

Cloudflare (Reverse Proxy)

AWS Application load balancer

EC2 (k3s worker node)

traefik ingress layer

backend service

frontend service

Backend
Server Pod

Backend
Server Pod

Frontend
Server Pod

Frontend
Server Pod

AWS Network Load Balancer

EC2 (k3s control plane)

API Server

etcd

scheduler

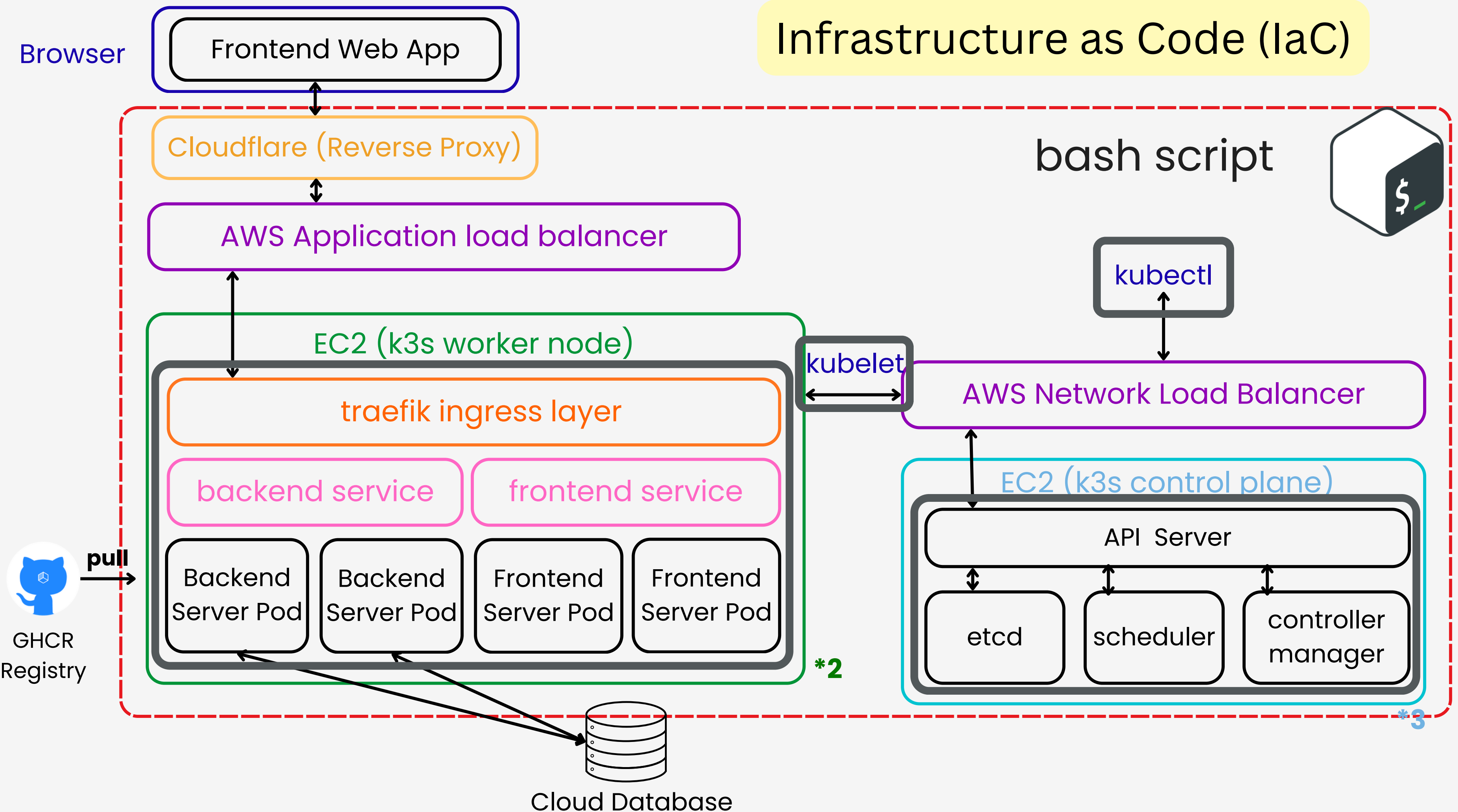
controller
manager

Cloud Database

*2

*3

Infrastructure as Code (IaC)



Infrastructure as Code (IaC)

完整部署僅需約 10 分鐘



kubectl

kubelet

AWS Network Load Balancer

EC2 (k3s control plane)

API Server

etcd

scheduler

controller manager

EC2 (k3s worker node)

traefik ingress layer

backend service

frontend service

Backend Server Pod

Backend Server Pod

Frontend Server Pod

Frontend Server Pod

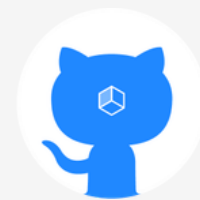
Cloudflare (Reverse Proxy)

AWS Application load balancer

Frontend Web App

Browser

pull



GHCR Registry

Cloud Database

*2

*3



系統自律

非功能面 Demo

Deployability & Maintainability

- Infrastructure as Code (IaC)
- Rolling Update

Scalability & Elasticity

- **Auto-scaling** : Real-time provisioning and deprovisioning

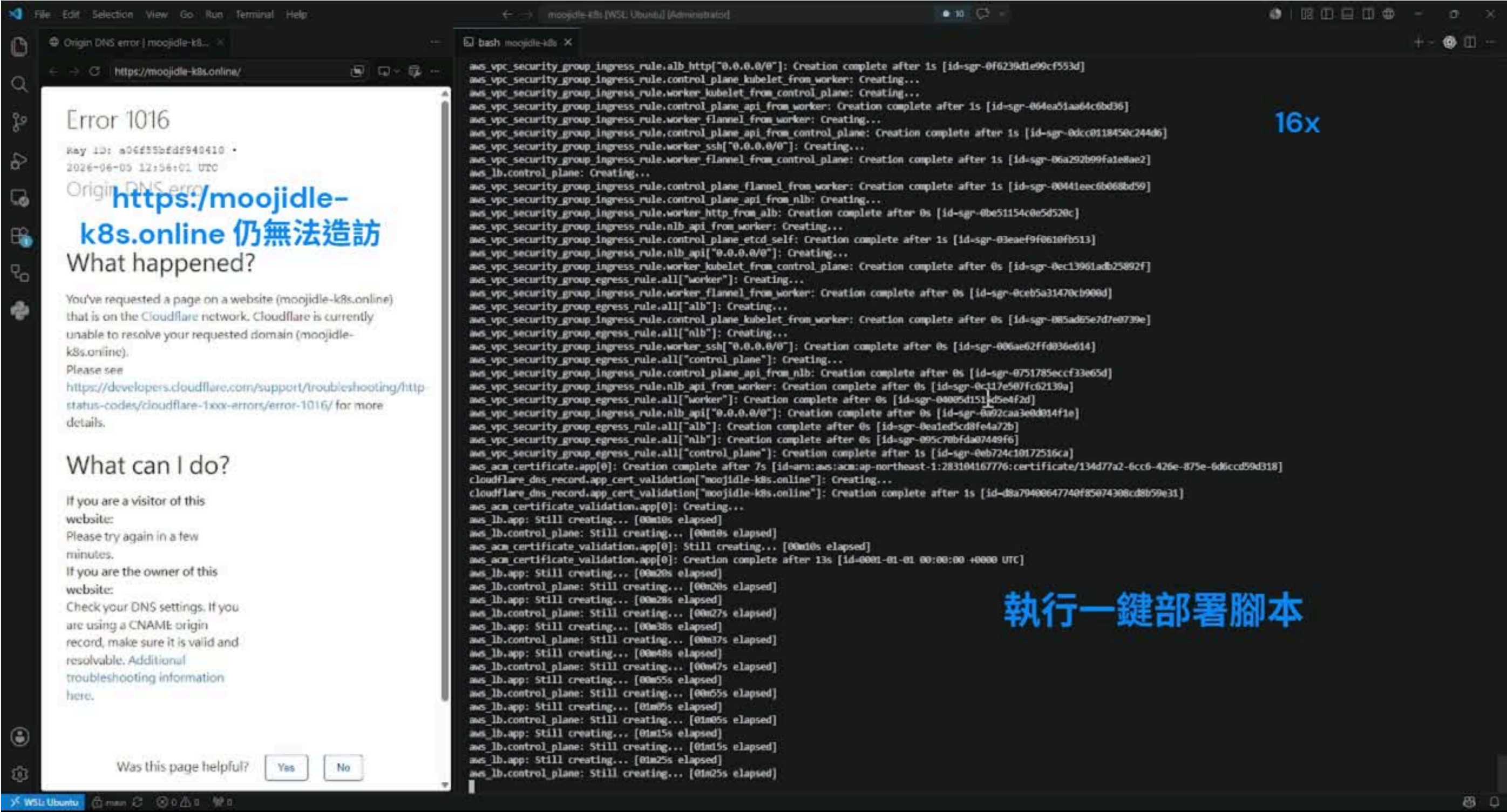
Observability

- Alerting

Reliability

- **Availability** —
 - Service (Pod)
 - Infrastructure (Node)
- **Recoverability (Fault Tolerance & Self-Healing)**

Infrastructure as Code (IaC)



Scalability & Elasticity

```
Every 1.0s: KUBECONFIG=~/.kube/moojidle-config kubectl get hpa      Jason: Wed Jun  3 19:02:32 2026
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
backend-hpa	Deployment/backend-deploy	cpu: 1%/70%	4	10	4	24m
frontend-hpa	Deployment/frontend-deploy	cpu: 4%/70%	2	5	2	24m

HPA (Horizontal Pod Autoscaler)

```
Every 1.0s: KUBECONFIG=~/.kube/moojidle-config kubectl get pods    Jason: Wed Jun  3 19:02:33 2026
```

NAME	READY	STATUS	RESTARTS	AGE
backend-deploy-86d75cd447-c6grw	1/1	Running	0	24m
backend-deploy-86d75cd447-crqkh	1/1	Running	0	24m
backend-deploy-86d75cd447-v2mx7	1/1	Running	0	24m
backend-deploy-86d75cd447-wgzcf	1/1	Running	0	24m
frontend-deploy-bd85f58b7-45gdz	1/1	Running	0	24m
frontend-deploy-bd85f58b7-n9vrz	1/1	Running	0	24m

Pod

```
Every 1.0s: KUBECONFIG=~/.kube/moojidle-config kubectl top pods ... Jason: Wed Jun  3 19:02:33 2026
```

POD	NAME	CPU(cores)	MEMORY(bytes)
backend-deploy-86d75cd447-c6grw	backend	1m	28Mi
backend-deploy-86d75cd447-crqkh	backend	1m	28Mi
backend-deploy-86d75cd447-v2mx7	backend	1m	27Mi
backend-deploy-86d75cd447-wgzcf	backend	1m	28Mi
frontend-deploy-bd85f58b7-45gdz	frontend	1m	3Mi
frontend-deploy-bd85f58b7-n9vrz	frontend	1m	3Mi

Pod CPU / Memory 用量

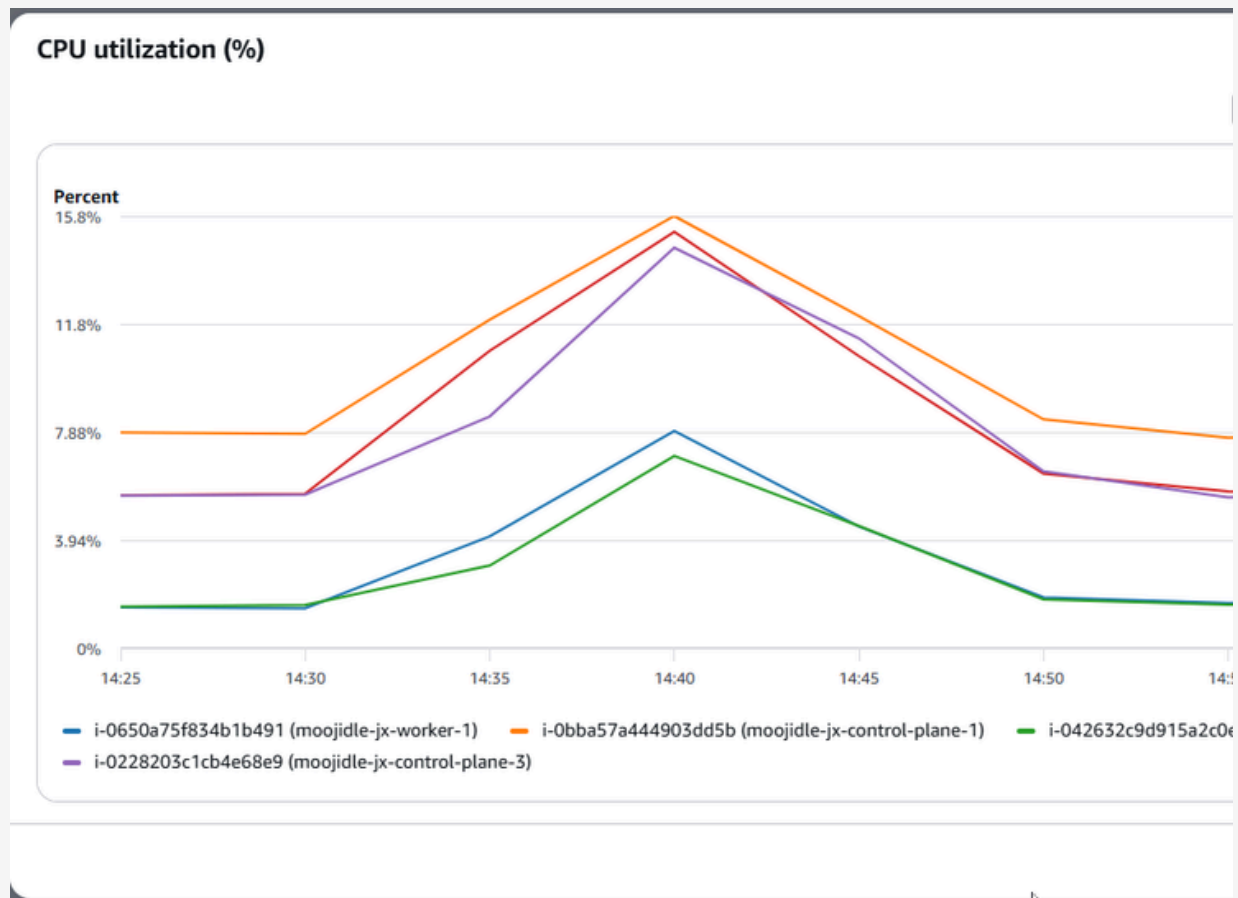
```
jason@Jason:~/moojidle-k8s$ BASE_URL=https://moojidle-k8s.online TARGET_VUS=200 k6 run load-tests/moojidle-stress.js
```

K6 壓測

VU: 200

30s / 150s / 30s

Observability (系統偵測)



使用K6壓測

VU: 200

30s / 9min / 30s

```
MOOJIDLE-K8S [WSL: UBU... Every 1.0s: KUBECTLCONFIG=~/kube/moojidle-config kubectl get hpa jx: Fri Jun 5 22:37:46 2026
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
backend-hpa	Deployment/backend-deploy	cpu: 25%/70%	4	10	4	92m
frontend-hpa	Deployment/frontend-deploy	cpu: 25%/70%	2	5	2	92m

```
MOOJIDLE-K8S [WSL: UBU... Every 1.0s: KUBECTLCONFIG=~/kube/moojidle-config kubectl get pods jx: Fri Jun 5 22:37:46 2026
```

NAME	READY	STATUS	RESTARTS	AGE
backend-deploy-54df6d7487-bwcnw	1/1	Running	0	31m
backend-deploy-54df6d7487-dnwdv	1/1	Running	0	92m
backend-deploy-54df6d7487-fv9rv	1/1	Running	0	92m
backend-deploy-54df6d7487-qmjm	1/1	Running	0	92m
frontend-deploy-bd85f58b7-sjkdg	1/1	Running	0	92m
frontend-deploy-bd85f58b7-znhzk	1/1	Running	0	92m

```
MOOJIDLE-K8S [WSL: UBU... Every 1.0s: KUBECTLCONFIG=~/kube/moojidle-config kubectl get pods jx: Fri Jun 5 22:37:46 2026
```

```
BASE_URL=https://moojidle-k8s.online/ TARGET_VUS=200 k6 run load-tests/moojidle-stress.js
```

```
execution: local
script: load-tests/moojidle-stress.js
output: -

scenarios: (100.00%) 1 scenario, 200 max VUs, 11m30s max duration (incl. graceful stop):
  * default: Up to 200 looping VUs for 11m0s over 3 stages (gracefulRampDown: 30s, gracefulStop: 30s)

running (00m26.8s), 195/200 VUs, 1984 complete and 0 interrupted iterations
default [>-----] 195/200 VUs 00m29.3s/11m00.0s
```

Reliability (Fault tolerance + Recoverability)

The screenshot displays a VS Code editor with a project named 'moojidle-k8s'. The Explorer sidebar on the left shows the file structure, including 'load-tests' and 'terraform' directories. The 'moojidle-stress.js' file is selected in the 'load-tests' directory.

The main editor area is split into two panes. The left pane shows the 'moojidle-stress.js' file with the following content:

```
load-tests > JS moojidle-stress.js > default
1 import http from 'k6/http';
```

The right pane shows the terminal output of the 'k6 run' command. The output includes the following tables:

Table 1: HPA Configuration

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
backend-hpa	Deployment/backend-deploy	cpu: 2%/70%	4	10	4	165m
frontend-hpa	Deployment/frontend-deploy	cpu: 4%/70%	2	5	2	165m

Table 2: Pod Status

NAME	READY	STATUS	RESTARTS	AGE
backend-deploy-54df6d7487-fv9rv	1/1	Running	0	165m
backend-deploy-54df6d7487-k875c	1/1	Running	0	165m
backend-deploy-54df6d7487-qm7zm	1/1	Running	0	165m
backend-deploy-54df6d7487-xvc59	1/1	Running	0	165m
frontend-deploy-bd85f58b7-sjkdg	1/1	Running	0	165m
frontend-deploy-bd85f58b7-zmhzk	1/1	Running	0	165m

The terminal output also shows the command 'k6 run load-tests/moojidle-stress.js' being executed, with the following output:

```
BASE_URL=https://moojidle-k8s.online/ TARGET_WUS=200 k6 run load-tests/moojidle-stress.js
```

Overlaid on the terminal output is the text:

K6 壓測
VU: 200
30s / 90s / 30s

Reliability (Fault tolerance + Recoverability)

The screenshot displays the AWS Management Console's EC2 Instances page. The left sidebar shows navigation options like Dashboard, AWS Global View, Events, and various instance types. The main content area shows a table of five instances. The instance 'moojidle-jx-control-plane-1' is selected, and its details are shown in the lower panel. The details panel includes sections for Instance summary, Public IPv4 address, Private IPv4 addresses, Public DNS, Private IP DNS name (IPv4 only), Instance type, VPC ID, Elastic IP addresses, and AWS Compute Optimizer finding.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6
moojidle-jx-worker-1	i-0650a75f834b1b491	Running	t3.small	3/3 checks passed	ap-northeast-1c	ec2-13-112-125-143.ap...	13.112.125.143	-	-
moojidle-jx-control-plane-1	i-0bba57a444903dd5b	Running	t3.medium	3/3 checks passed	ap-northeast-1c	ec2-13-158-4-101.ap-n...	13.158.4.101	-	-
moojidle-jx-worker-2	i-042632c9d915a2c0e	Running	t3.small	3/3 checks passed	ap-northeast-1a	ec2-3-112-215-176.ap...	3.112.215.176	-	-
moojidle-jx-control-plane-2	i-0b31e37f572b4ea76	Running	t3.medium	3/3 checks passed	ap-northeast-1a	ec2-13-192-204-107.ap...	13.192.204.107	-	-
moojidle-jx-control-plane-3	i-0228203c1cb4e68e9	Running	t3.medium	3/3 checks passed	ap-northeast-1d	ec2-35-77-37-221.ap-n...	35.77.37.221	-	-

**於 AWS Console 手動停止
1 個 Control Plane
1 個 Worker Node**

Instance summary

Instance ID: i-0bba57a444903dd5b

IPv6 address: -

Hostname type: IP name: ip-172-31-1-191.ap-northeast-1.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: -

Public IPv4 address: 13.158.4.101 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-1-191.ap-northeast-1.compute.internal

Instance type: t3.medium

VPC ID: vpc-02308b9b587e6f70a

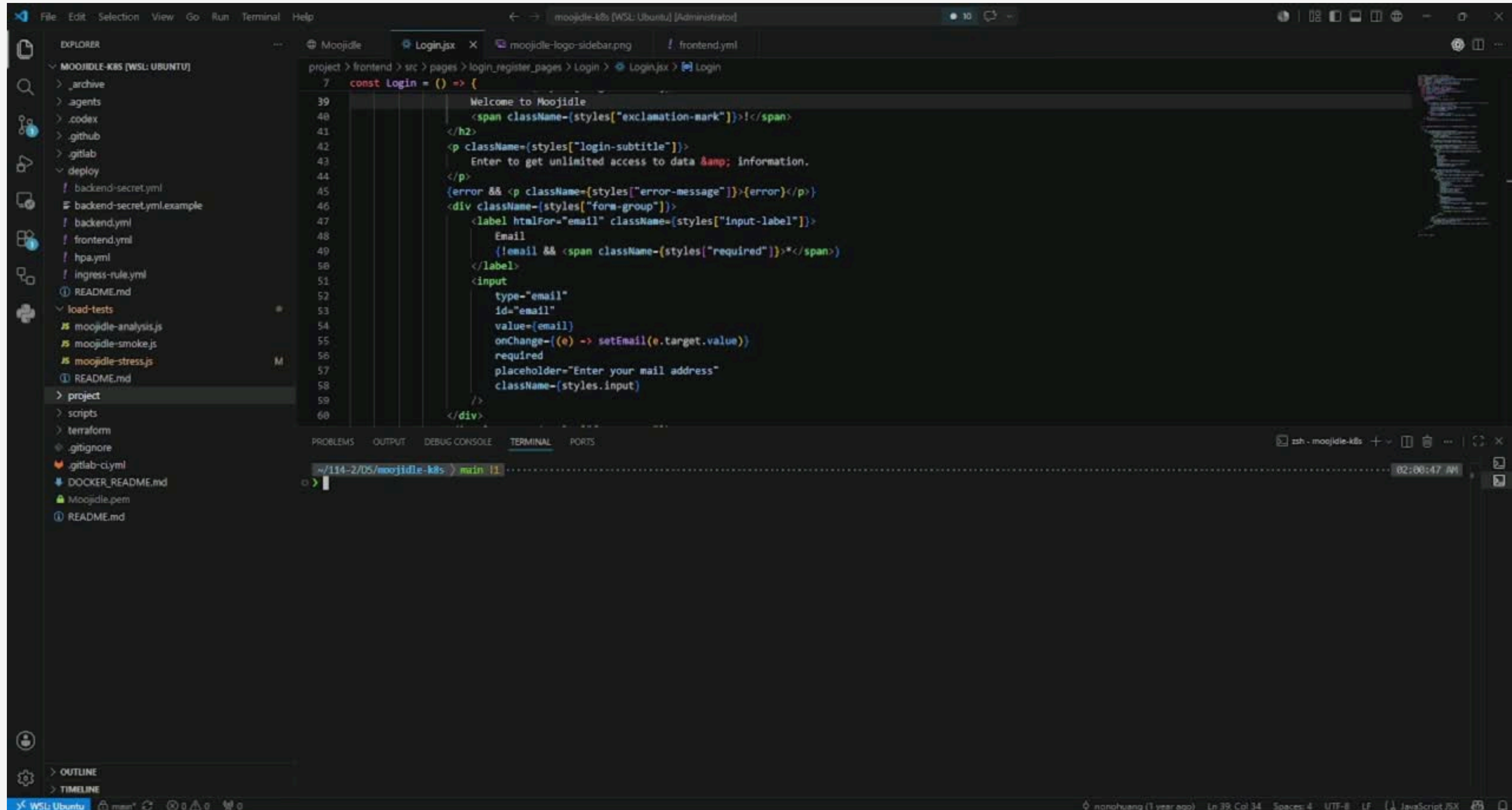
Private IPv4 addresses: 172.31.1.191

Public DNS: ec2-13-158-4-101.ap-northeast-1.compute.amazonaws.com | open address

Elastic IP addresses: -

AWS Compute Optimizer finding: -

Deployability (rolling update)



THANK YOU
FOR LISTENING

完整 DEMO 連結

https://www.youtube.com/watch?v=-B0SvC_Mhl8

Github 連結

<https://github.com/Jason9339/moojidle-k8s>